Quiz

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Main notions learned in this chapter

5/5 points (graded)

The solution to the analogical equation: A is to B as C is to X is the value of X that has minimal complexity.

- ( ) True
- (•) False

✔

Answer
Correct: False. It should minimize the overall complexity of (A,B,C,X).

Unsupervised prototype-based clustering achieves compression because distances to prototypes require less information.

- (•) True
- ( ) False

✔

Answer
Correct:
True. Prototypes are chosen so that distances from data points to prototypes are smaller and require fewer bits to be encoded at a given precision.

Outliers in clustering are data points with low-complexity coordinates.

- ( ) True
- (•) False

✔

Answer
Correct:
False. Outliers are indeed simple (if there are few of them), but this is because their coordinates remain complex despite the clustering.

Trained neural networks invove millions, sometimes billions of parameter values. Yet, the point of using neural networks is still to achieve data compression.

- (•) True
- ( ) False

✔

Answer
Correct:
True. They are used to perform prediction (e.g. in classification task). Compression is achieved whenever the dataset used after training is large enough.

MDL achieves lossless compression.

- (•) True
- ( ) False

Answer
Correct: True. In (crude) MDL, all exceptions are described individually.

Submit You have used 1 of 1 attempt

Correct (5/5 points)

Problem 1

2/2 points (graded)

Astronomers collected 128 observational data about an exoplanet in a double star system. A standard hypothesis E_1 about the mass of the planet and its distance to the stars accounts for m of the 128 data points. The $(128 - m)$ exceptions remain a mystery, unless the two stars follow a particular (never observed) elongated elliptic trajectory around each other (hypothesis E_2). If the two stars happen to obey E_2, it accounts for all data. However, the precision of the additional parameter is found to add 48 bits to the complexity of E_1 alone.

For which value of m does $E_1 + E_2$ offer a better theory than E_1 alone?

122

122

Answer
Correct:
The cost of the exceptions to E_1 is $(128 - m) \times \log_{2+}(128) = 7 \times (128 - m)$ bits.
This cost has to be larger than E_2's 48 bits for E_2 to be useful. The number m of data explained by E_1 alone should be smaller than: $m \leq 128 - 48/7) \approx 121$ for $E_1 + E_2$ to be preferred.

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Correct (2/2 points)

Problem 2

0/2 points (graded)

A shop's database D contains data about clients. They are distinguished by their age A and the amount S in Euros (no cents) they spent in the shop during the period. Some clients spent nearly as much as 500€.

A learning system detected a strong positive correlation σ between age and the amount spent. As a result, data points (A, S) are well predicted by $\hat{S} = \sigma A + s_0$. The difference between S and $\hat{S}$ is found to never exceed $\pm$30€.

Knowing the regression model $M = (\sigma, s_0)$, the learning system compresses the initial data D down to $K(D|M)$.

What is the amplitude of this compression %b(per data point)?

16 - 12 = 4 bits per data point

Answer
Incorrect:
For each data point, we need a few bits to represent age (7 bits for age up to 128 years) plus 9 bits to represent the amount spent (up to 512€).

Thanks to the correlation, we just need to represent age plus 5 bits for the prediction error (which is smaller than 30€) plus one bit to indicate the sign of the error.
So the compression is 3 bits per data point.

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✖ Incorrect (0/2 points)

Problem 3

0/2 points (graded)

A dataset includes 4000 points in a 4-D space. Values along each of the four dimensions are positive and no larger than 1000.
Divide each value range in 4. This delimits 256 regions of the 4D-cube of size 1000. A clustering technique found that data are located in 16 clusters included within the 16 corner-regions of the cube.

What is the overall compression obtained through the clustering?

160 - 112 = 48 kbit compression ✖

Answer
Incorrect:
Initially, we need 4000 x 4 x log2(1000) = 160 kbits to locate the data.
After the clustering, we need only one bit per axis to locate a cluster. We need 4000 x 4 x (1 + log2(250)) = 144 kbits.
So the compression is about 16 kbits. This amounts to 4 bits per item.

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✖ Incorrect (0/2 points)

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